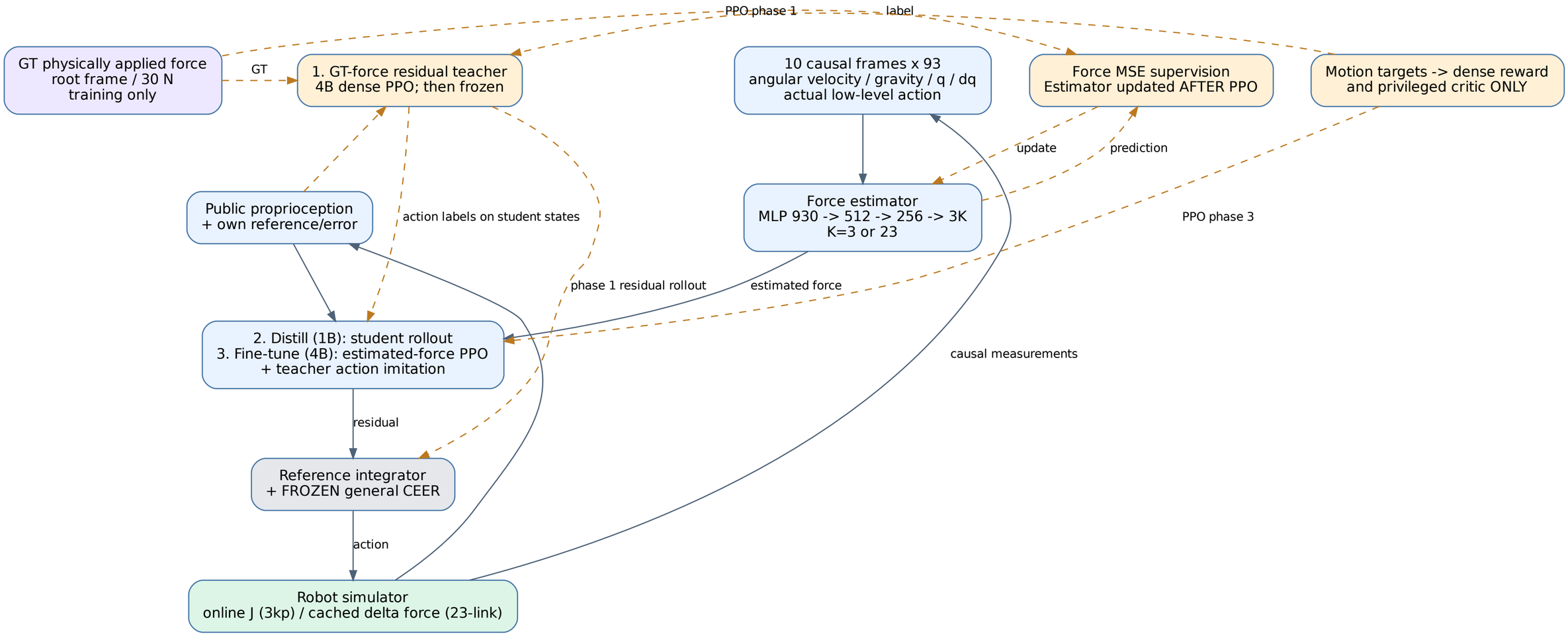E | Force-estimated residual student: three training phases